

since probab "k" would have in degree = k_b $= p(k_b, t) = \frac{n(k_b, t)}{N_b(t)}$

(2) $\Rightarrow [(k+1)p(k_b, k+1) - k p(k_b, k)] (1-g)$

$= \tilde{Z}^{-1} [(\lambda_b(k_b+1) + \mu_b) p(k_b+1, t) - (\lambda_b k_b + \mu_b) p(k_b, t)] (1-g) + (1-g) \delta_{k_b, 0}$

As $t \rightarrow \infty$ $p(k_b, t) = p_{\infty}(k_b)$:

$\Rightarrow \text{LHS} : p_{\infty}(k_b) = \tilde{Z}^{-1} [(\lambda_b(k_b+1) + \mu_b) p_{\infty}(k_b+1) - (\lambda_b k_b + \mu_b) p_{\infty}(k_b)] + \delta_{k_b, 0} \quad (3)$

if $k_b > 0 \Rightarrow \delta_{k_b, 0} = 0$

$\Rightarrow p_{\infty}(k_b) [\tilde{Z}^{-1} (\lambda_b k_b + \mu_b + 1)] = p_{\infty}(k_b+1) [\tilde{Z}^{-1} (\lambda_b k_b + (\mu_b - \lambda_b))]$

$\Rightarrow \frac{p_{\infty}(k_b)}{p_{\infty}(k_b+1)} = \frac{k_b + \mu_{k_b} + \frac{\lambda_b}{\tilde{Z}}}{k_b + \mu_{k_b} + 1 + \frac{\lambda_b}{\tilde{Z}}} = 1$

$\Rightarrow p_{\infty}(k_b) = A_b \frac{\Gamma(k_b + \mu_{k_b})}{\Gamma(k_b + \mu_{k_b})}$

for $k_b \geq 0 : p_{\infty}(k_b=0) \Rightarrow (3) \Rightarrow p_{\infty}(0) = \tilde{Z}^{-1} [-\mu_b p_{\infty}(0)] + \delta_{0,0}$

$\Rightarrow p_{\infty}(0) (1 + \mu_b \tilde{Z}^{-1}) = 1 \Rightarrow p_{\infty}(0) = (1 + \mu_b \tilde{Z}^{-1})^{-1}$

using (3) from $k > 0$:

$\therefore p_{\infty}(0) = (1 + \mu_b \tilde{Z}^{-1})^{-1} \cdot 1 = A \frac{\Gamma(\mu_{k_b} + 1 + \frac{\lambda_b}{\tilde{Z}})}{\Gamma(\mu_{k_b})}$

$\Rightarrow A_b = \frac{\Gamma(\mu_{k_b})}{\Gamma(\mu_{k_b} + 1 + \frac{\lambda_b}{\tilde{Z}})} \cdot (1 + \mu_b \tilde{Z}^{-1}) \cdot 1$

conjecture by problem symmetry $\Rightarrow$

$p(k_a^{(n)}) = A_a \frac{\Gamma(\mu_{k_a} + 1 + \frac{\lambda_a}{\tilde{Z}})}{\Gamma(\mu_{k_a})}$

$A_a = \frac{\Gamma(\mu_{k_a})}{\Gamma(\mu_{k_a} + 1 + \frac{\lambda_a}{\tilde{Z}})} \cdot (1 + \mu_a \tilde{Z}^{-1})$